

Project Initialization and Planning Phase

Date	02 October 2026
Team ID	SWTID-2026-5625
Project Title	FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks	

Project Proposal (Proposed Solution) Report

This proposal aims to simplify personal fitness planning using generative AI. FitBuddy creates personalized 7-day workout and food plans from basic user details. Key features include Gemini-powered plan generation, health-aware recommendations, and feedback-based plan updates.

Project Overview	
Objective	Deliver personalized, easy-to-follow fitness and nutrition plans using Google Gemini AI, with no need for a trainer or dietitian.
Scope	A web application that collects the user's name, age, weight, fitness goal, workout intensity and optional health information. It generates a 7-day workout plan, a 7-day food plan, nutrition and recovery tips, and updates the plan based on user feedback.
Problem Statement	
Description	Generic fitness plans ignore individual goals, body details and health conditions. Personal trainers and dietitians are costly and not accessible to everyone.
Impact	Users get affordable, personalized and health-aware guidance, which improves consistency, safety and motivation.
Proposed Solution	
Approach	Use Google Gemini models through a FastAPI backend to turn user inputs into structured plans, shown through a simple Jinja2/HTML interface.
Key Features	- Personalized 7-day workout plan - Personalized 7-day food plan - Nutrition and recovery tips - Health-aware workout considerations - Feedback-based plan updates - Stored user data and plans (SQLite)

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		

Computing Resources	CPU (AI runs on Gemini cloud API; no GPU needed)	Intel i3/i5 or equivalent
Memory	RAM specifications	8 GB
Storage	Disk space for code, database and logs	256 GB SSD
Software		
Frameworks	Python frameworks	FastAPI, Uvicorn
Libraries	Additional libraries	jinja2, sqlalchemy, python-multipart, google-genai, python-dotenv, pydantic
Development Environment	IDE / tools	Visual Studio Code, Git, GitHub
Data		
Data	Source, size, format	User-entered inputs and AI-generated plans stored in SQLite; Gemini API as AI source

Team Members and Roles

Name	Role
Sharmila D	Team Lead
Bhavana A	Backend Developer
Rajashree K	AI/GenAI Developer
Siva Sundari V	Frontend/UI Developer
Praveena K	Development and Template Developer